UNIVERSITY OF WAH(UOW)

DEPARTMENT OF COMPUTER SCIENCE

Lab Task # 03

COURSE: Data Structure & Algorithm Lab_201L

SUBMITTED TO: Ms.Muniba Khan

SUBMITTED BY: Muhammad Talha_CS.3ndB

Registration NO: UW-24-CS-BS-101

Task # 01

```cpp
int main() {
    int numbers[10];
    cout << "Enter 10 integers:" << endl;
    for (int i = 0; i < 10; i++)
    {
        cout << "Number " << i + 1 << ": ";
        cin >> numbers[i];
    }
    cout << "\nNumbers you entered are: ";
    for (int i = 0; i < 10; i++)
     {
        cout << numbers[i] << " ";
    }
    cout << "\nEven numbers are: ";
    for (int i = 0; i < 10; i++)
    {
        if (numbers[i] % 2 == 0)
        {
            cout << numbers[i] << " ";
        }
    }
    cout << endl;
```

Output

```
Enter 10 integers:
Number 1: 1
Number 2: 2
Number 3: 3
Number 4: 4
Number 5: 5
Number 6: 6
Number 7: 7
Number 8: 8
Number 9: 9
Number 10: 11

Numbers you entered are: 1 2 3 4 5 6 7 8 9 11
Even numbers are: 2 4 6 8
```

Task # 02

```cpp
int main() {
    int a[3][3];
    int i, j;
    cout << "Enter elements of 3x3 matrix:" << endl;
    for (i = 0; i < 3; i++) {
        for (j = 0; j < 3; j++) {
            cin >> a[i][j];
        }
    }
    cout << endl << "Matrix:" << endl;
    for (i = 0; i < 3; i++) {
        for (j = 0; j < 3; j++) {
            cout << a[i][j] << "\t";
        }
        cout << endl;
    }
    cout << endl << "Sum of each row:" << endl;
    for (i = 0; i < 3; i++) {
        int sum = 0;
        for (j = 0; j < 3; j++) {
            sum += a[i][j];
        }
        cout << "Row " << i + 1 << " = " << sum << endl;
    }
```

```cpp
cout << endl << "Sum of each column:" << endl;
for (j = 0; j < 3; j++) {
    int sum = 0;
    for (i = 0; i < 3; i++) {
        sum += a[i][j];
    }
    cout << "Column " << j + 1 << " = " << sum << endl;
}
cout << endl << "Main diagonal elements: ";
for (i = 0; i < 3; i++) {
    cout << a[i][i] << " ";
}
cout << endl;
return 0;
```

Output

```
Enter elements of 3x3 matrix:
1 3 4
54 7 9
2 54 22

Matrix:
1       3       4
54      7       9
2       54      22

Sum of each row:
Row 1 = 8
Row 2 = 70
Row 3 = 78

Sum of each column:
Column 1 = 57
Column 2 = 64
Column 3 = 35

Main diagonal elements: 1 7 22
```

Task # 03

```cpp
int main() {
    int n, i;
    cout << "Enter a number: ";
    cin >> n;
    cout << endl << "Multiplication table using for loop:" << endl;
    for (i = 1; i <= 10; i++) {
        cout << n << " x " << i << " = " << n * i << endl;
    }
    cout << endl << "Multiplication table using while loop:" << endl;
    i = 1;
    while (i <= 10) {
        cout << n << " x " << i << " = " << n * i << endl;
        i++;
    }
    cout << endl << "Multiplication table using do-while loop:" << endl;
    i = 1;
    do {
        cout << n << " x " << i << " = " << n * i << endl;
        i++;
    } while (i <= 10);
    return 0;
```

Output

```
Multiplication table using while loop:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Task # 04

```cpp
void getNumbers(int a[]) {
    cout << "Enter 10 numbers:" << endl;
    for (int i = 0; i < 10; i++)
        cin >> a[i];
}

void countNumbers(int a[], int &even, int &odd, int &zero) {
    even = odd = zero = 0;
    for (int i = 0; i < 10; i++) {
        if (a[i] == 0)
            zero++;
        else if (a[i] % 2 == 0)
            even++;
        else
            odd++;
    }
}

void showResult(int even, int odd, int zero) {
    cout << endl << "Even = " << even << endl;
    cout << "Odd = " << odd << endl;
    cout << "Zero = " << zero << endl;
}

int main() {
    int a[10], even, odd, zero;
    getNumbers(a);
    countNumbers(a, even, odd, zero);
    showResult(even, odd, zero);
```

Output

```
Enter 10 numbers:
12
3
44
5
7
1
99
110
11
87

Even = 3
Odd = 7
Zero = 0
```

Task # 05

```cpp
int n;
cout << "Enter how many numbers: ";
cin >> n;
int a[n];
cout << "Enter " << n << " numbers:" << endl;
for (int i = 0; i < n; i++)
    cin >> a[i];
cout << endl << "Reversed array:" << endl;
for (int i = n - 1; i >= 0; i--)
    cout << a[i] << " ";
cout << endl;
```

Output

```
Enter how many numbers: 5
Enter 5 numbers:
5 4 3 2 1

Reversed array:
1 2 3 4 5
```

Task # 06

```cpp
int main() {
    int a[5];
    cout << "Enter 5 numbers:" << endl;
    for (int i = 0; i < 5; i++)
        cin >> a[i];
    cout << endl << "Numbers entered: ";
    for (int i = 0; i < 5; i++)
        cout << a[i] << " ";
    cout << endl;
    int max = a[0];
    int min = a[0];
    for (int i = 1; i < 5; i++) {
        if (a[i] > max)
            max = a[i];
        if (a[i] < min)
            min = a[i];
    }
    int diff = max - min;
    cout << "Maximum = " << max << endl;
    cout << "Minimum = " << min << endl;
    cout << "Difference = " << diff << endl;
    return 0;
```

Output

```
Enter 5 numbers:
4 3 2 6 8

Numbers entered: 4 3 2 6 8
Maximum = 8
Minimum = 2
Difference = 6
```

Task # 07

```cpp
int factorial(int n) {
    int fact = 1;
    for (int i = 1; i <= n; i++)
        fact *= i;
    return fact;
}

int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;

    cout << "Factorial of " << n << " = " << factorial(n) << endl;
    return 0;
```

Output

```
Enter a number: 5
Factorial of 5 = 120
```

Task # 08

```
int main() {
    int n;
    cout << "Enter number of elements: ";
    cin >> n;

    int a[n];
    int totalSize = sizeof(a);
    int elementSize = sizeof(a[0]);
    int fit = 100 / elementSize;

    cout << endl << "Total size of array = " << totalSize << " bytes" << endl;
    cout << "Each element size = " << elementSize << " bytes" << endl;
    cout << "Elements that can fit in 100 bytes = " << fit << endl;

    return 0;
```

Output

```
Enter number of elements: 5

Total size of array = 20 bytes
Each element size = 4 bytes
Elements that can fit in 100 bytes = 25
```